Orthographic projection

National 5 · Drawing & spatial awareness · Worksheet

GRAPHICS WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ identify and arrange elevation, plan and end elevation in third-angle projection
- ☐ show hidden detail correctly in every view
- ☐ work out missing sizes from an orthographic drawing
- ☐ identify true lengths and true shapes

I DO — Worked example

Read how a full-mark answer is built

An L-shaped bracket is shown as a pictorial view. Four sets of orthographic views are shown, labelled A to D.

1 State what you would check to decide which set of views is correctly arranged in third-angle projection. (2 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ The plan is drawn above the elevation
- ✓ The end elevation is on the side it is viewed from

WE DO — Guided practice

Use the starters and word bank to help you

A shelf bracket has an overall length of 200. From the left it has a 62 flange, then a slot, then a 72 end section.

2 Calculate the length of the slot. (1 mark)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 1 marks = 1 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) In third-angle projection, the plan is drawn…

☐ Above the elevation ☐ Below the elevation ☐ To the left of the elevation ☐ On a separate sheet

b) Hidden detail is shown with…

☐ Thick continuous lines ☐ Thin dashed lines ☐ Chain lines ☐ Wavy lines

c) To find the true shape of a sloping cut you look…

☐ From the front elevation ☐ Looking down from directly above ☐ From the end elevation ☐ At 90° to the cut surface

d) The third-angle symbol shows a cone frustum with its circles…
☐ On the left ☐ Above ☐ On the right ☐ Below

4 Key terms: write the letter of the matching description next to each term.

1. **Third-angle projection: where is the plan?**	____	**A** Above the elevation.
2. **Elevation**	____	**B** The main front view of an object.
3. **End elevation**	____	**C** Views lined up so sizes project between them.
4. **Related views**	____	**D** The side view, drawn on the side it is viewed from in third angle.

5 Explain what is meant by ‘Hidden detail’. (2 marks)

6 Explain what is meant by ‘True shape’. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?